THE CO-OPERATIVE UNIVERSITY OF KENYA

UNIT CODE: BCSC 3209 TITLE: ARTIFICIAL INTELLIGENCE (A.I)

ASSIGNMENT ONE

INSTRUCTIONS:

- **Attempt all the questions in GROUPS OF 2-5 STUDENTS. No individual responses will be accepted**
- **Submit your answers by 2nd April, 2026 in PDF format Submit answers via eMasomo (NO email submission shall be accepted).**
- **Every student in the group must submit the document independently indicating group members on the answer response.**

QUESTIONS:

a) Consider the following information given by consumers surveying bureau of Kenya. Use the symbolic representation to represent the rules as used in production-rule knowledge representation. (8 marks)

> "if one lives near Muthaiga road and has a plump salary then he is likely to join Muthaiga golf club.
> One can say he leaves near Muthaiga road if he leaves less than 15 kilometers off the road.
> The survey also reveals that you have to have a plump income and be a who-is-who in Nairobi so that you can join an exclusive club.
> If you can join an exclusive club, then you are likely to meet politicians.
> One who meets politicians is said to be well connected.
> If one can join exclusive and is a who-is-who in Nairobi, then he is well connected.
> If well connected, then one can join Muthaiga golf club.
> If one can join Muthaiga golf club then he is likely to meet the president."

b) Artificial Intelligence (AI) currently encompasses a huge variety of subfields, ranging from general-purpose areas, such as learning and perception to such specific tasks as playing chess, proving mathematical theorems, writing poetry, and diagnosing diseases. It systematizes and automates intellectual tasks and is therefore potentially relevant to any sphere of human intellectual activity. In this sense, it is truly a universal field. Unlike many existing fields like Physics, Biology and Chemistry, AI still presents various open challenges and research problems. Discuss any five specialization areas of Artificial Intelligence. (10 marks)

c) You are hired as knowledge based systems' engineer to develop a medical diagnosis system for Afya Bora Medical Services. In your task, you are expected to provide a detailed functionality expectation of your intelligent agent. Further, you are expected to provide an explanation to the board on the nature of environment the system will operate and interact with the existing systems. Using this background, answer the following questions;

i. Using the PEAS framework, explain the expectations of the Agent. (3 Marks)
ii. Using appropriate illustrations, explain two probable operation environments that the system can operate within (5 Marks)
iii. Write a detailed pseudo-code for the main functionality of the system. (4 Marks)